Kondo and RKKY interactions in bilayer graphene

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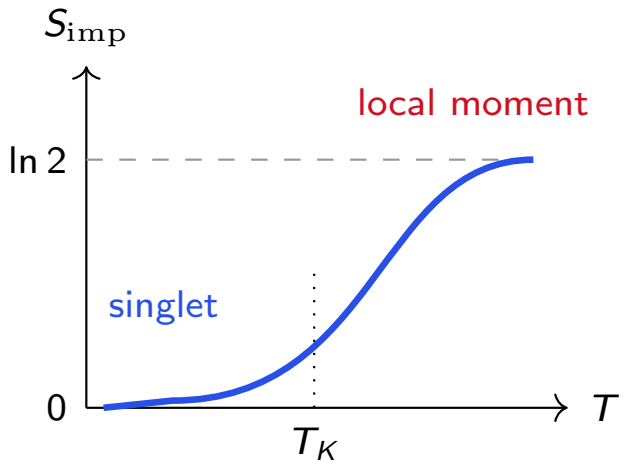
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NM4QM 2026

A Kondo impurity screens at T_K, and the density of states sets T_K



$S_{\rm imp}$: $\ln 2 \to 0$ as the moment is screened

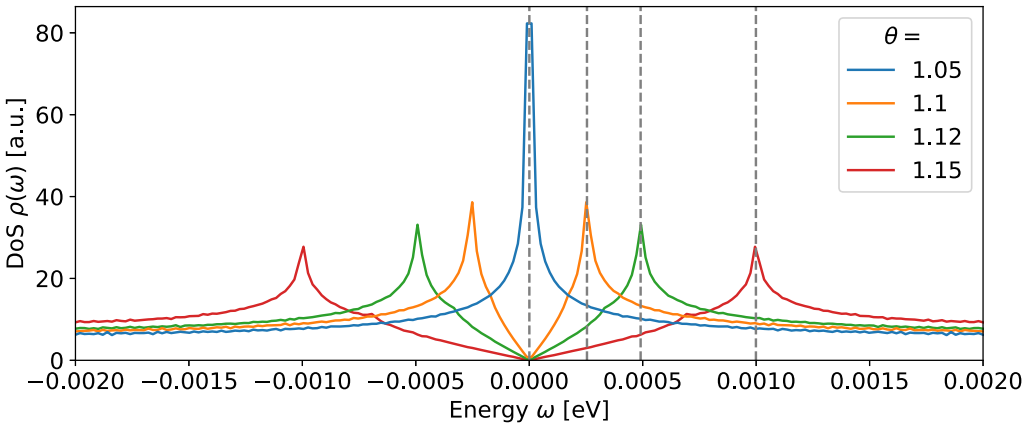
$T_K \sim D\, e^{-1/\rho_0 J}$: larger DoS, larger T_K

Singular DoS: enhanced, or suppressed, T_K

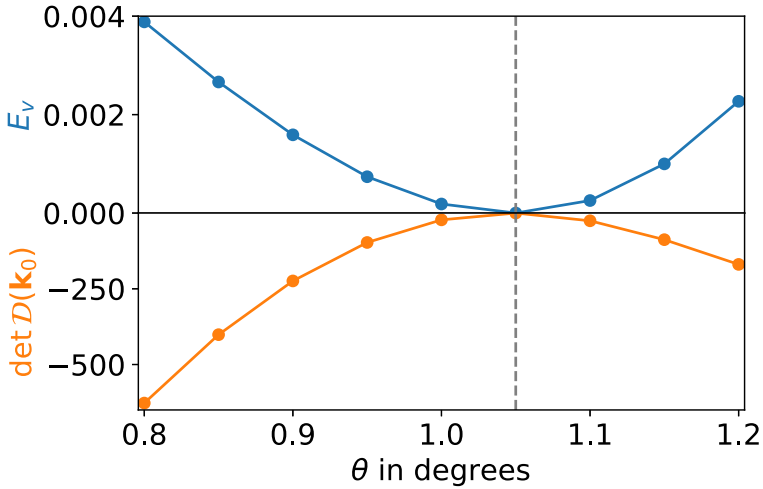
	Density of states, $\rho(\omega)$	Kondo temperature, T_K	Impurity entropy, $S_{\rm imp}(T=0)$
Metal	ρ_0 : constant	$De^{-1/\rho_0 J}$	0
Standard van Hove	$\rho_0[1 + a\ln(D/\lvert\omega\rvert)]$: logarithmic diverging	$De^{\frac{1}{a}-\frac{1}{a}\sqrt{1+\frac{2a}{\rho_0 J}}}$	0
Higher Order van Hove	$\rho_0\lvert\omega\rvert^{-\alpha}$: power-law diverging	$D(1+\frac{\alpha D^\alpha}{\rho_0 J})^{-1/\alpha}$	$-2\alpha\ln 2$
Dirac cone	$\rho_0\lvert\omega\rvert$: linear pseudogap	$T_K = 0$	$\ln 2$

Entropy $S_{\rm imp}(T)$ is the single diagnostic used throughout

Twisted bilayer graphene: two van Hove singularities merge at 1.05°



(b)



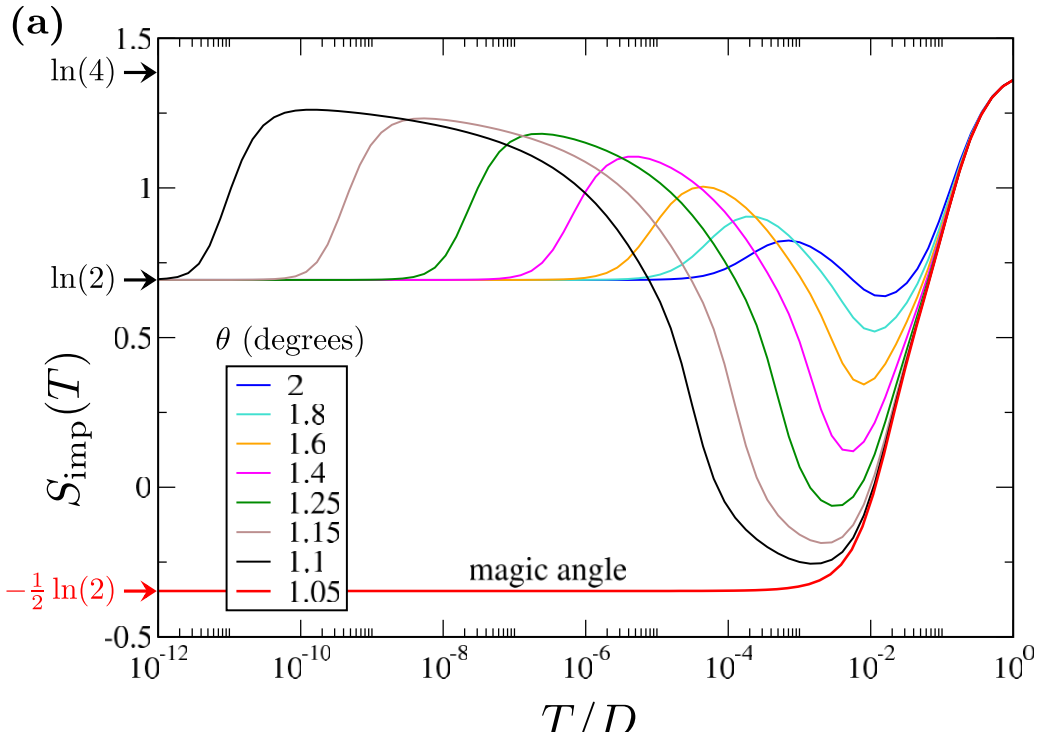
(c)

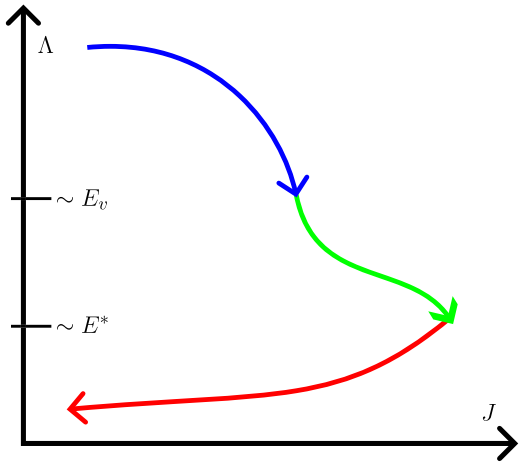
Saddle-point energies E_s collapse onto one another as $\theta \to \theta_{\text{magic}}$

Higher-order vHs: power-law divergent DoS

Impurity physics tracks the host DoS as a function of twist angle

At the magic angle the entropy overshoots and the moment is restored





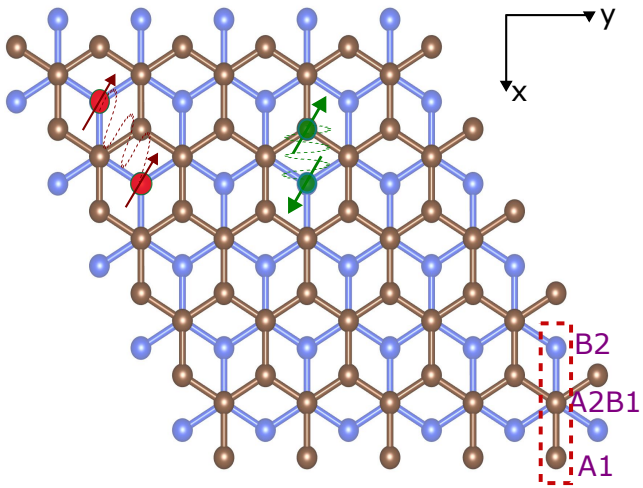
$S_{\rm imp} \to -\frac{1}{2}\ln 2$ at $\theta = 1.05^\circ$

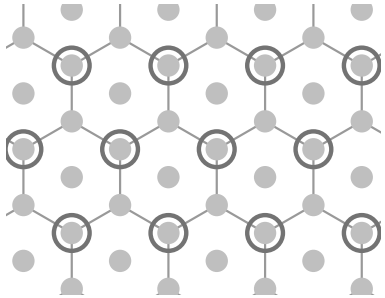
RG: HO-vHs $\to$ log-vHs $\to$ Dirac

$\ln 4$ overshoot, free moment below E^*

Next: no twist needed, the impurity *position* is the knob (bilayer graphene)

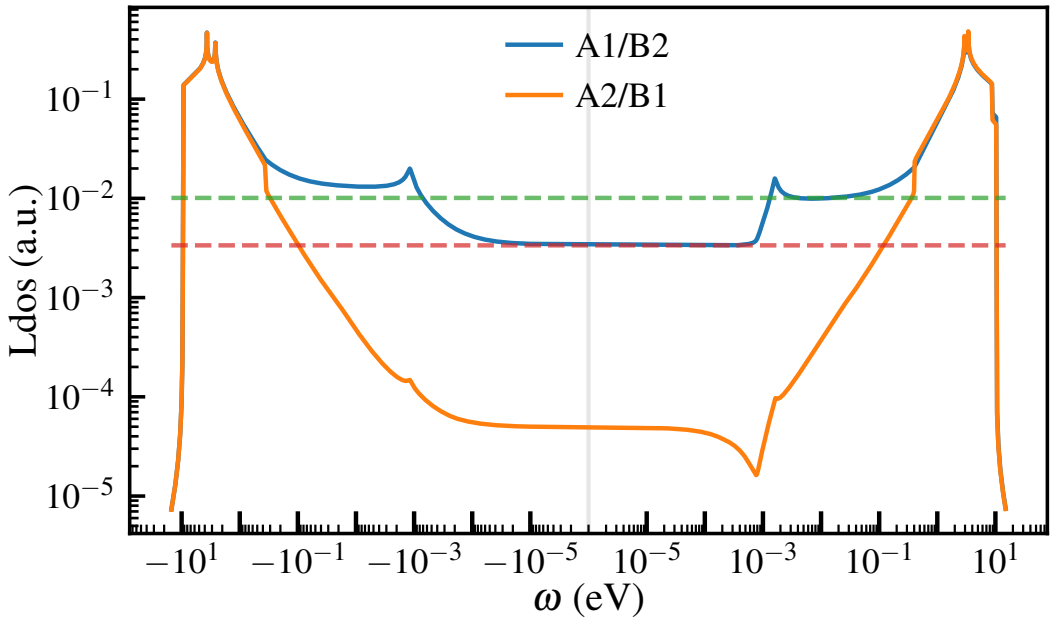
Bernal bilayer: stacked and unstacked sites see very different DoS





unstacked: A1, B2

stacked: A2/B1

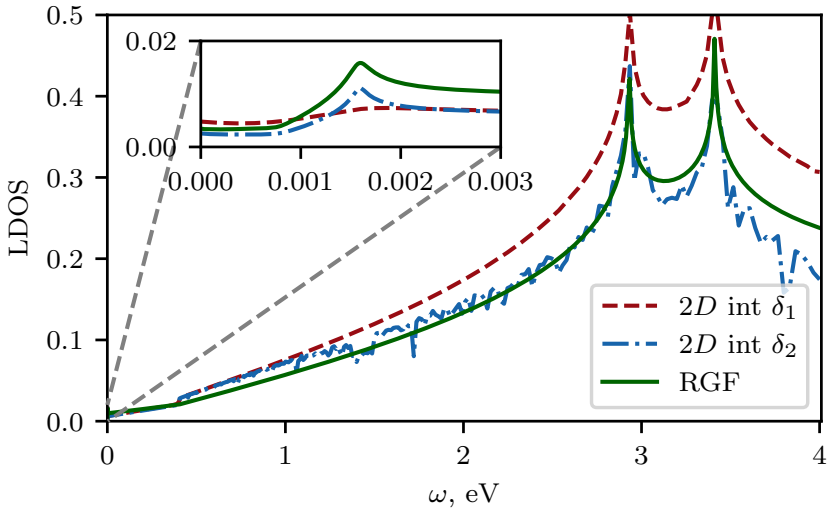


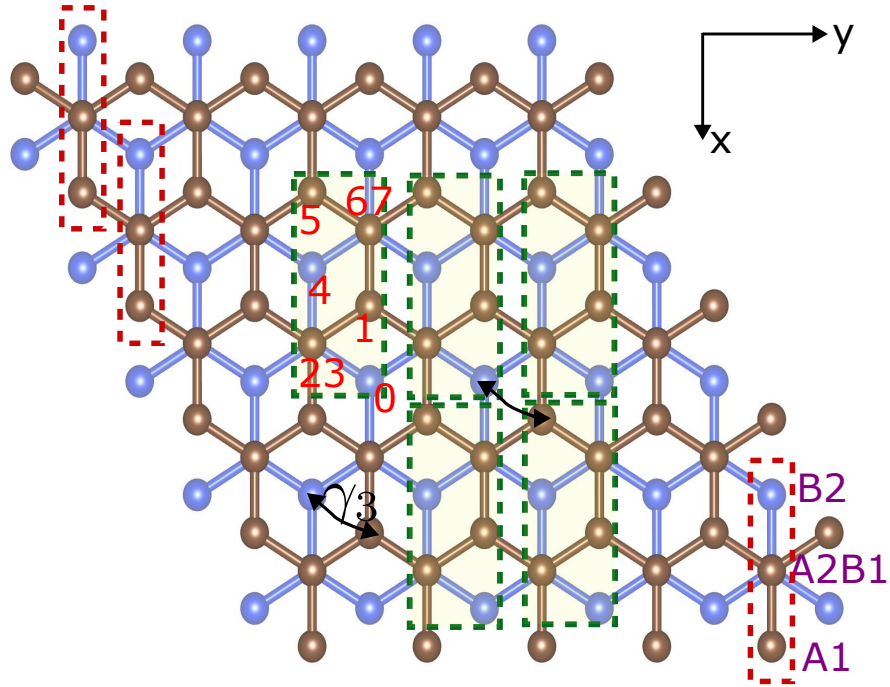
A1/B2 (unstacked): large LDoS, trigonal-warping vHs at $\sim$1.6 meV

A2/B1 (stacked): suppressed LDoS at low energy

Same lattice, two kinds of site: what does an impurity feel?

RGF resolves the 1.6 meV vHs that 2D integration misses



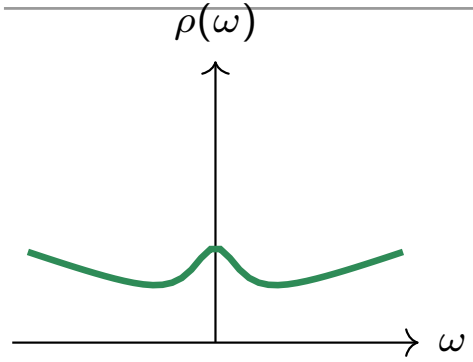


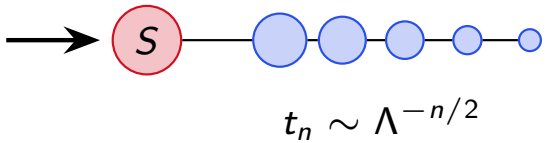
vHs lives within 10^{-4} of the BZ scale: 2D integration needs a huge grid

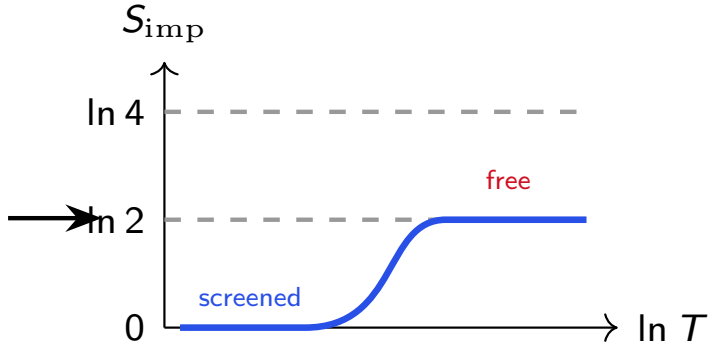
Fold into a square supercell, Fourier transform along one axis, recurse

RGF gives the full energy-resolved local and non-local DoS cheaply

NRG turns the DoS into entropy plateaus, each one a fixed point







1. bath DoS (RGF) **2.** Wilson chain, $S \cdot \sigma$ at the end **3.** $S(T)$, $\chi(\omega)$

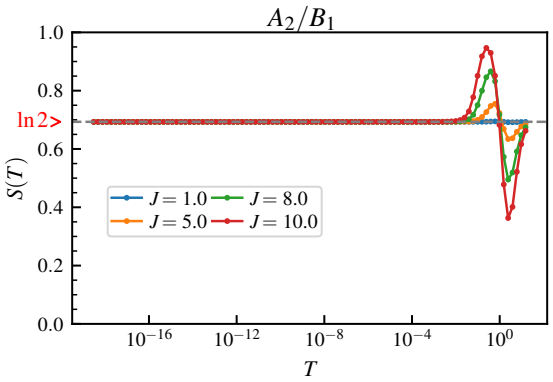
Plateau value identifies the fixed point: free moment ($\ln 2$), singlet (0), spin-1 ($\ln 3$)

Peak of $-\mathrm{Im}\,\chi(\omega)$ marks the screening scale

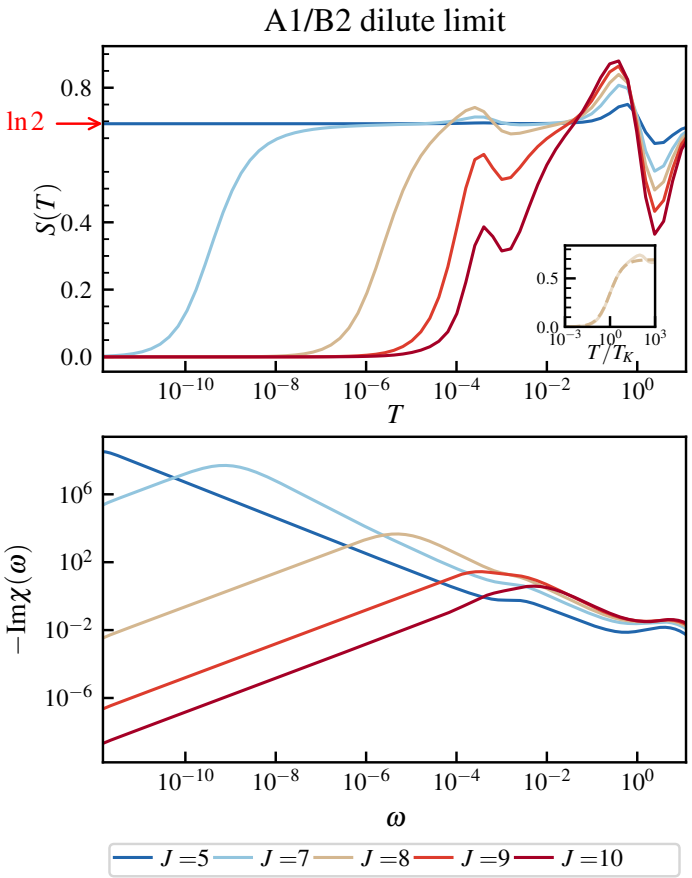
We read the physics off $S_{\rm imp}(T)$, no model-specific fits needed

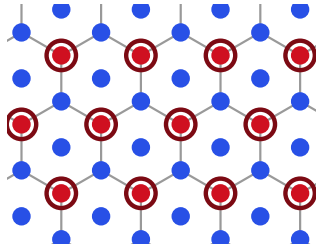
Stacked sites never screen; unstacked sites do

A2/B1



A1/B2





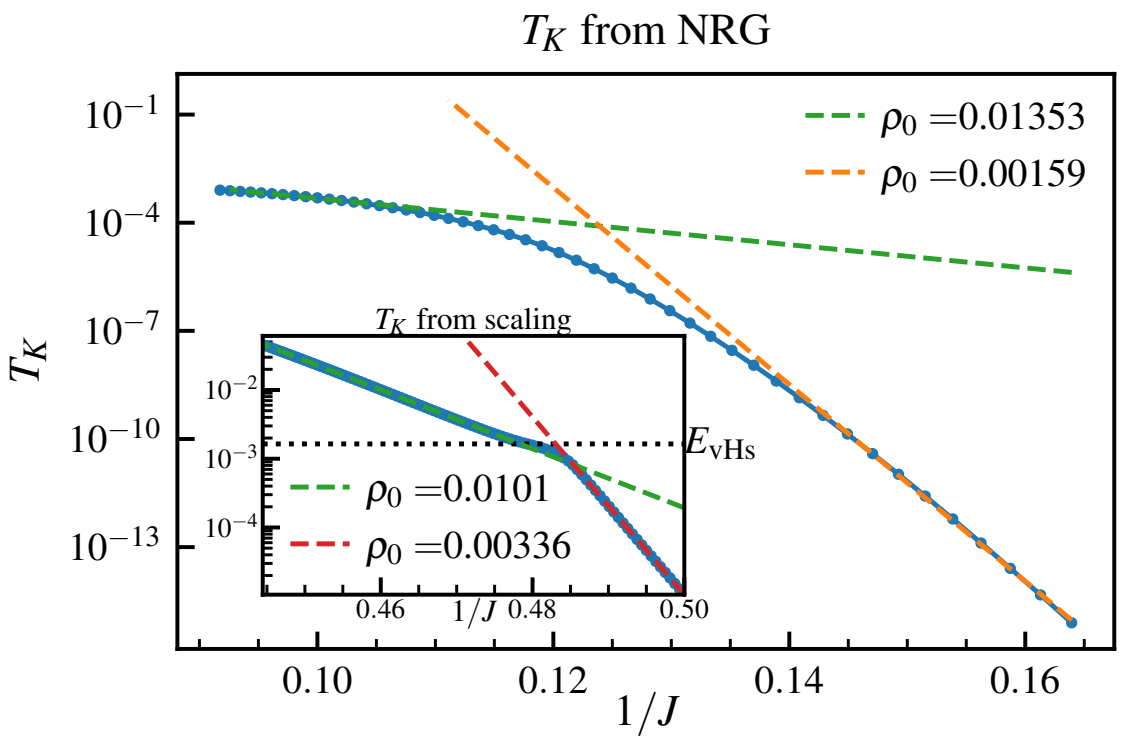
- free moment (A2/B1)
- Kondo (A1/B2)

A2/B1: $S \to \ln 2$

A1/B2: $S \to 0$ at large J

Free moment vs Kondo singlet is set by the impurity position alone

T_K crosses between two DoS plateaus as the coupling grows



Weak J: low-energy DoS plateau

Strong J: flow reaches the high-energy plateau

Smooth crossover, no transition

$$T_K \sim D \exp\left(-1/\rho_{\mathrm{eff}} J\right)$$

Poor man's scaling captures it qualitatively; NRG confirms it quantitatively

Two impurities: the RKKY sign follows the sublattice

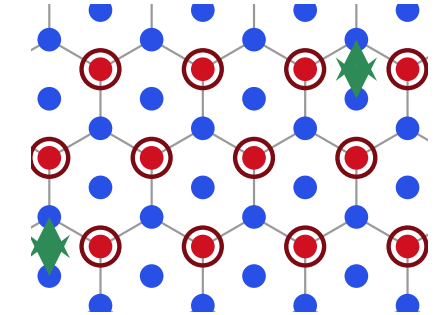
$$H = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}} + J \sum_{i=1,2} \mathbf{S}_i \cdot \boldsymbol{\sigma}(\mathbf{r}_i)$$

Same sublattice: **FM**

Opposite sublattice: **AFM**

$J_{\mathrm{RKKY}} \propto J^2$, $T_K \sim e^{-1/J}$

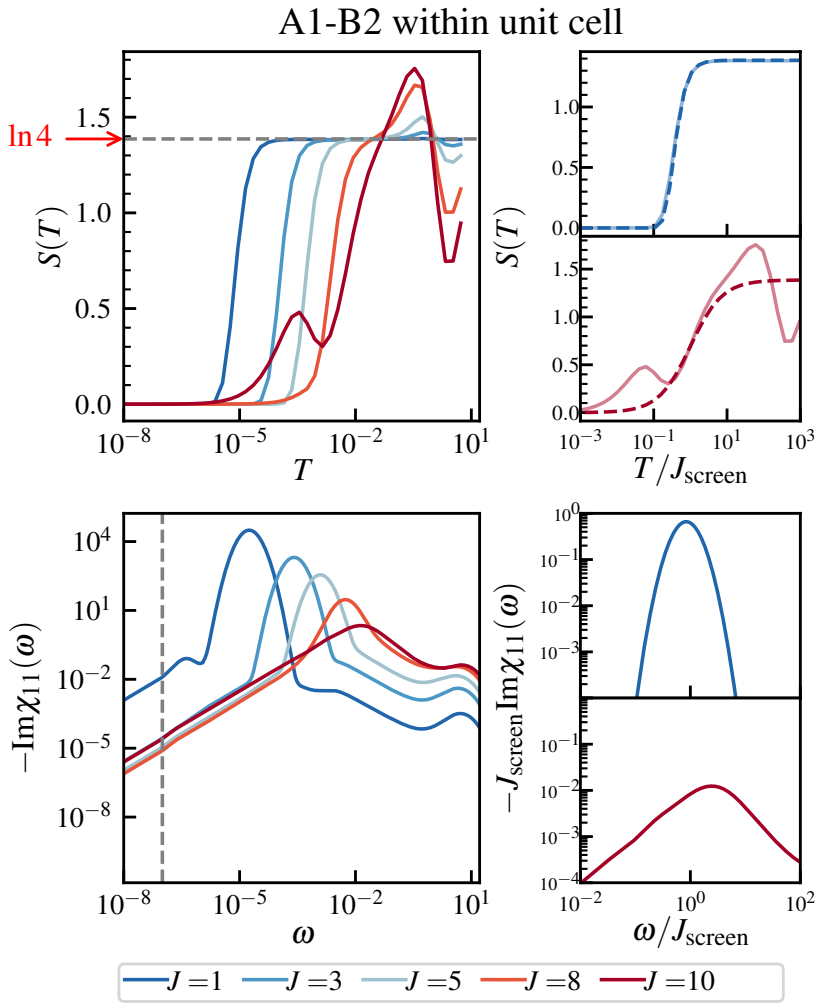
pair	competing regimes
A2–A2	FM / LM
A2–B1	AFM / LM
A1–B2	AFM / Kondo
B2–B2	FM / Kondo



- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)

Only A1–B2 and B2–B2 have a nontrivial RKKY–Kondo competition

A1–B2: an AFM singlet at weak J crosses over to Kondo screening



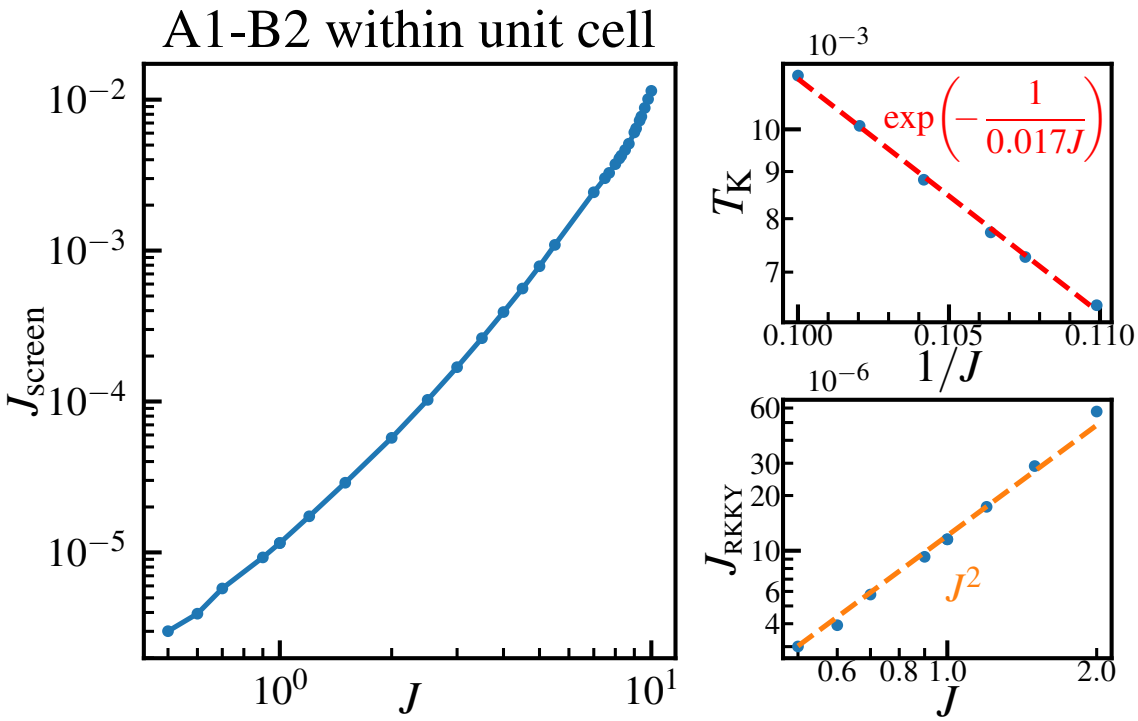
Weak J: entropy $\ln 4 \to 0$ via $\ln 2$, effective two-spin model $J_{\mathrm{RKKY}}\mathbf{S}_1\cdot\mathbf{S}_2$

Strong J: two independent Kondo screenings

Both limits follow universal curves

Antiferromagnetic locking and Kondo screening compete at one scale

The screening scale goes from J^2 (RKKY) to $e^{-1/J}$ (Kondo)



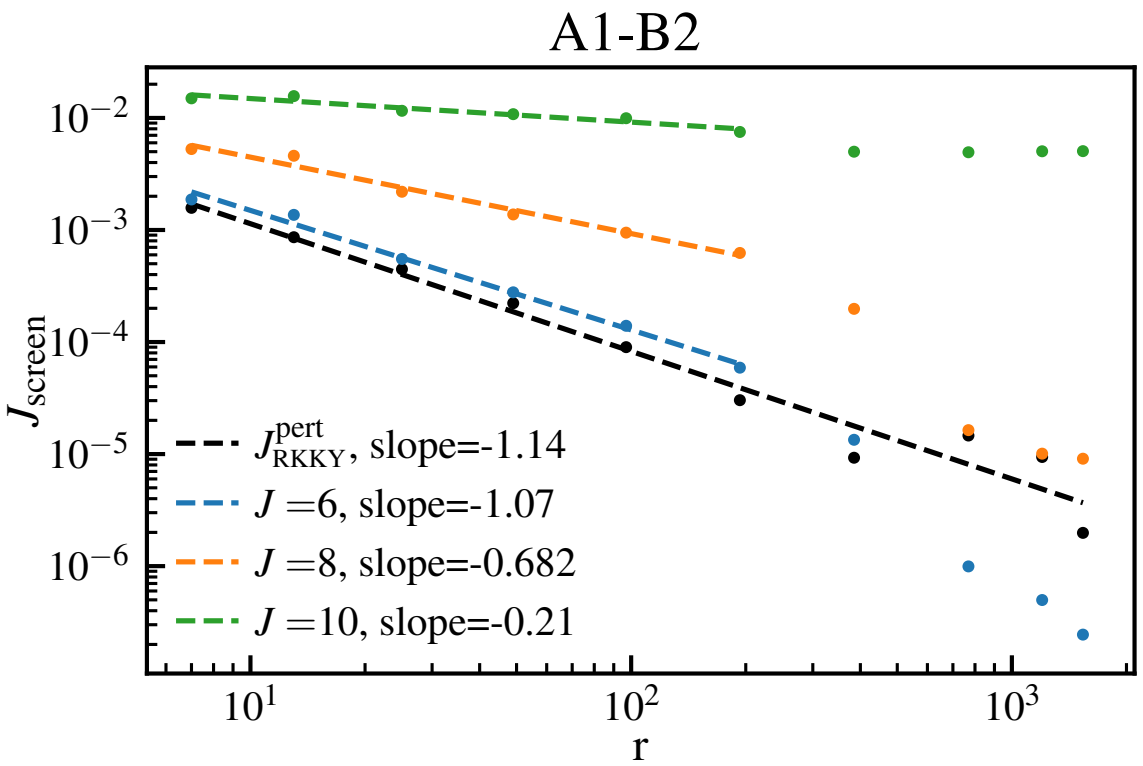
Weak J: $J_{\mathrm{screen}} \propto J^2$

Strong J: $\propto e^{-1/J}$

Insets confirm both laws

Same configuration, two regimes: the competition is non-perturbative

The distance dependence flattens from -1.14 to -0.21 with coupling



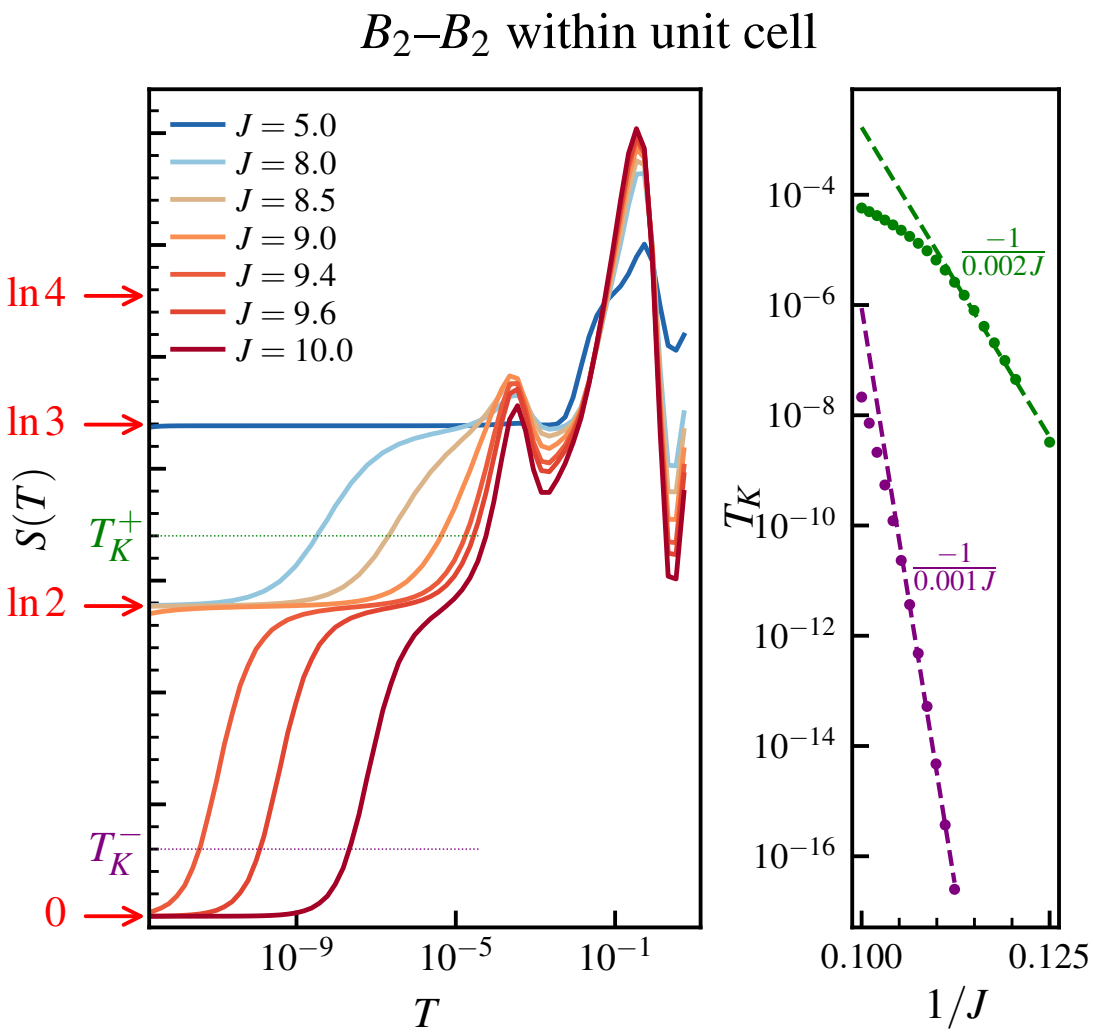
Perturbative RKKY: slope -1.14

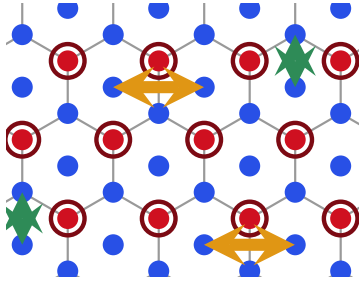
$J = 10$: slope -0.21

Strong coupling approaches the dilute Kondo limit

At strong coupling the impurities barely feel each other

B2–B2: an FM lock leads to two-stage Kondo screening





- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

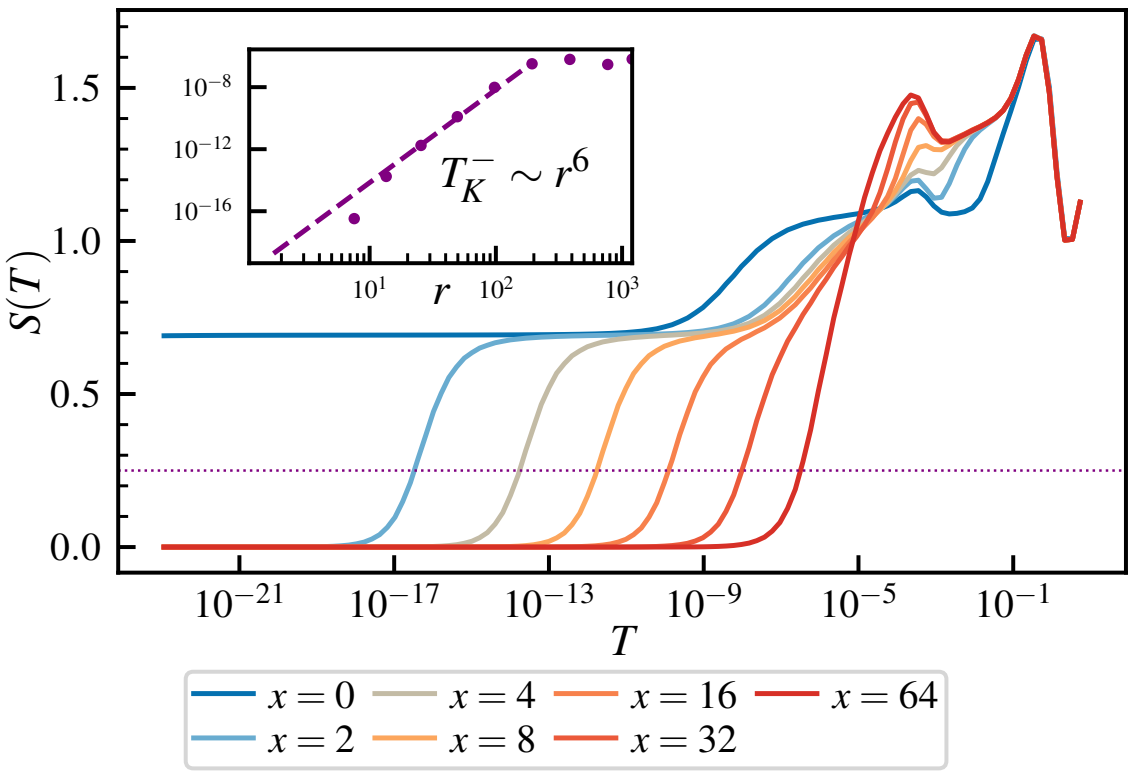
$\ln 4 \to \ln 3$ (FM spin-1)

$\to \ln 2$ (underscreened, T_K^+)

$\to 0$ (singlet, T_K^-)

Entropy plateaus: $\ln 4 \to \ln 3 \to \ln 2 \to 0$

With distance the two Kondo scales merge into the dilute limit



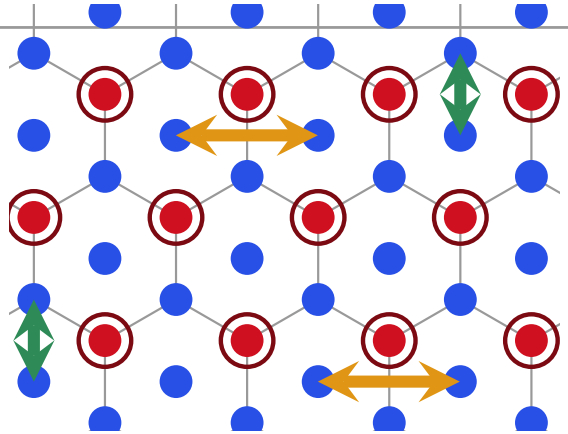
$T_K^{\pm}$ approach each other as r grows

Two-stage structure needs a sizeable FM RKKY

Dilute limit recovered smoothly

Two-stage screening is strongest for close impurities

One host, every screening regime: the impurity position selects it



- free moment (A2/B1)
- Kondo (A1/B2)
- AFM singlet (A1–B2)
- FM two-stage (B2–B2)

Stacked: free moment, never screened

Unstacked: Kondo, AFM singlet, FM two-stage

Position and pairing pick the regime

Outlook: STM on BLG, artificial impurity arrays, RKKY in TBG

Site selectivity: every Kondo regime in one sample

Backup: universal two-spin entropy curves

Antiferromagnetic, $J > 0$: $S(J{=}T) = \ln 3 + \ldots$ (universal function of $u = J/4T$)

Ferromagnetic, $J < 0$: spin-1 plateau, $S \to \ln 3$

Kondo: $S(T) \simeq \ln 2/(1 + T_K/T)$, with $S(T_K) = \frac{1}{2} \ln 2$

Backup: perturbative RKKY, sign from the NLDOS

$$\rho(\mathbf{r},\omega) = \rho_s(\mathbf{r},\omega) + \rho_a(\mathbf{r},\omega), \quad \rho_{s/a}(-\omega) = \pm\rho_{s/a}(\omega)$$

$J_{\mathrm{RKKY}} \propto J^2$ from the one-loop spin susceptibility

Positive denominator: sign set by the symmetric vs antisymmetric NLDOS

Same sublattice: symmetric dominates, FM; opposite: AFM

Backup: toy model, RKKY develops a logarithmic decay

